

# FINAL REPORT - PNEUMONIA DETECTION

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## 1. Summary of problem statement, data and findings

### 1.1 Problem Statements

The primary objective of this project was to develop an automated system for the accurate detection and localization of pneumonia from chest X-ray images. This is a critical task in medical diagnostics, as timely and accurate detection of pneumonia is essential for patient care. The solution was designed to assist radiologists by providing a reliable initial screening tool.

The dataset used consisted of chest X-ray images, each labeled for the presence of pneumonia. The problem was approached in two distinct parts:

- **Image Classification:** Classifying an entire image as either "Pneumonia" or "Normal."
- **Object Detection:** Localizing areas of interest (lung opacities) with bounding boxes.

### 1.2 Dataset

The project used the RSNA Pneumonia Detection Challenge dataset, comprising over 30,000 chest X-rays in DICOM format. Images were annotated with labels indicating pneumonia presence and bounding-box annotations were provided for cases with lung opacities.

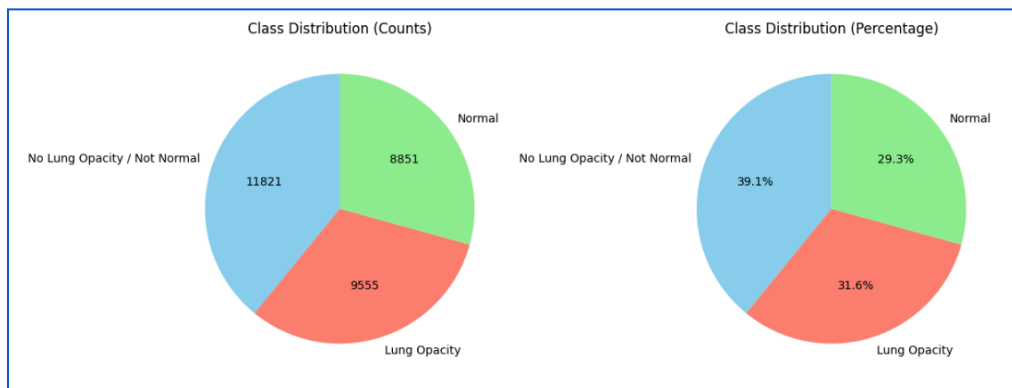
#### Class Distribution:

- Normal: 8,851 (~28.7%)
- Lung Opacity (Pneumonia): 9,555 (~31.0%)
- No Lung Opacity but Abnormal: 11,821 (~38.3%) (represents abnormal cases that may mimic pneumonia)

These detailed annotations allowed the project to address both classification and localization tasks.

| Categories                   | Class Distribution (Provided) |
|------------------------------|-------------------------------|
| Normal                       | 8,851 (~28.7%)                |
| Lung Opacity (Pneumonia)     | 9,555 (~31.0%)                |
| No Lung Opacity / Not Normal | 11,821 (~38.3%)               |

\* No Lung Opacity / Not Normal: abnormal but not pneumonia, sometimes mimicking pneumonia).



### 1.3 Milestone 2: Key Objectives

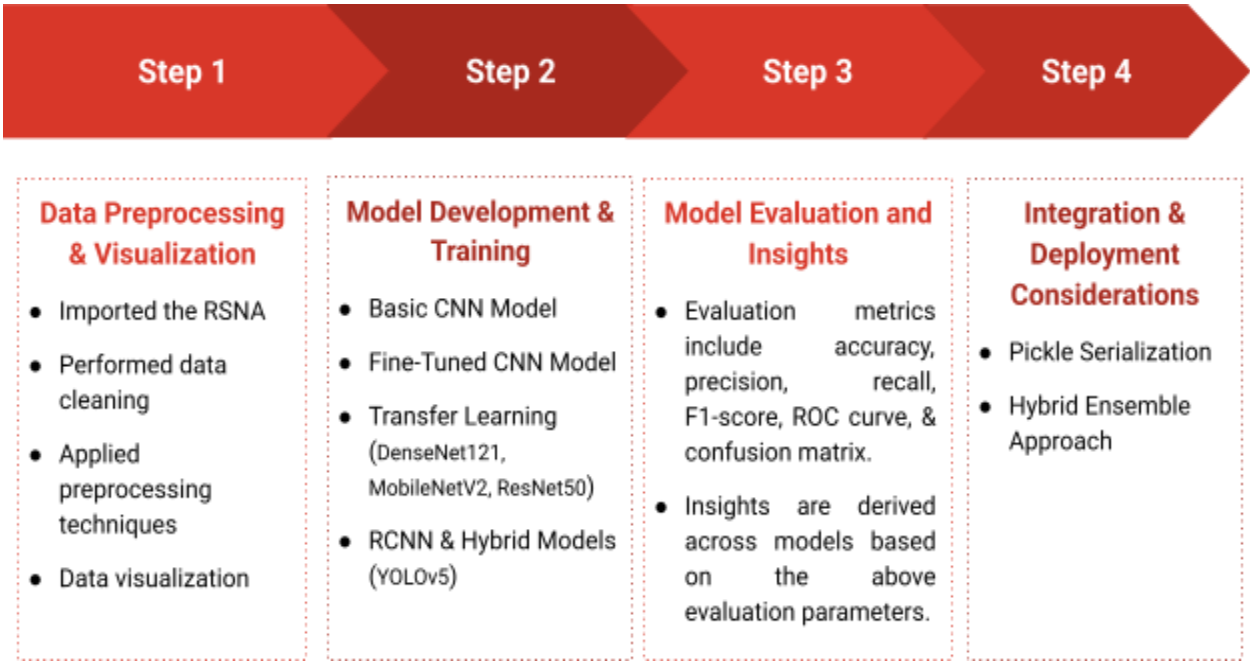
- **Data Preprocessing:** Standardized DICOM conversion to PNG, resized images, applied augmentation (flips, rotations, CLAHE).
- **Modeling:** Implemented and compared multiple architectures → Base CNN, Fine-Tuned CNN, ResNet50, DenseNet121, MobileNetV2, and YOLOv5 (for detection).
- **Evaluation Metrics:** Accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, ROC curves, and precision-recall curves.
- **Key Insight:** DenseNet121 achieved the best recall (0.47) and F1-score (0.53), making it most reliable for detecting pneumonia cases. MobileNetV2 achieved the highest precision (0.71) and ROC-AUC (0.83), excelling at avoiding false alarms.

### 1.4 Milestone 2: Summary of Findings

- **Classification Models:** DenseNet121 outperformed others in recall and F1-score, critical for minimizing false negatives in medical use cases. MobileNetV2 showed the best precision but lower recall, useful in ensembles.
- **Detection Model (YOLOv5):** Successfully localized pneumonia in 4/7 test images, but confidence levels were low (0.27–0.43), highlighting the need for further refinement.
- **Overall Conclusion:** Deep learning models can assist radiologists by flagging potential pneumonia cases. DenseNet121 is recommended as the primary screening model, with YOLOv5 needing further improvement for localization.

## 2. Overview of the final process

The pneumonia detection system was developed through a systematic, milestone-driven pipeline that balanced **classification accuracy** with **localization capability**. Each stage progressively built on the previous one, integrating **data preprocessing**, **advanced deep learning architectures**, **transfer learning techniques**, and **region-based object detection models**. The process ultimately produced a deployable, interpretable solution that demonstrates both diagnostic relevance and practical feasibility.



### 2.1 Data Preprocessing & Visualization

The foundation of the system was the RSNA Pneumonia Detection Challenge dataset, containing over 30,000 chest radiographs in DICOM format with corresponding metadata. The metadata included patient IDs, diagnostic labels (Normal, Lung Opacity, and No Lung Opacity/Not Normal), and bounding box annotations for pneumonia regions. Imported the RSNA dataset and mapped patient IDs with corresponding class labels and bounding-box annotations.

Key steps included:

- **Dataset Integration:** Patient IDs were mapped to class labels and bounding-box annotations, ensuring a consistent link between CSV metadata and DICOM images.

- **Data Cleaning:** Removed corrupted files, resolved mismatched metadata, and standardized class labels.
- **Image Preprocessing:**
  - Resized images to 224×224 pixels for model input.
  - Normalized intensity values to the 0–1 range.
  - Converted grayscale images to RGB channels to align with ImageNet-pretrained networks.
  - Applied augmentation techniques (rotation, flipping, zooming, brightness adjustment) to improve model generalization and reduce overfitting.
- **Visualization:**
  - Explored class distribution to assess imbalance across categories.
  - Validated bounding-box annotations against images.
  - Highlighted the challenge of visual overlap between Lung Opacity and Not Normal cases, which increases the likelihood of misclassification.

## 2.2 Model Development & Training

Model development followed a phased approach, moving from **baseline CNNs** to **transfer learning architectures** and **object detection frameworks**.

- **Basic CNN Model:** Designed, trained, and evaluated simple CNN architectures to establish initial performance benchmarks.
- **Fine-Tuned CNN Model:** The baseline CNN was fine-tuned using different hyperparameter adjustments and custom augmentation strategies.
- **Transfer Learning:** Leveraged pre-trained models (e.g., DenseNet121, ResNet50, MobileNetV2) to enhance classification accuracy by utilizing learned features from large-scale datasets.
  - **DenseNet121** emerged as the best performer, achieving the highest recall (0.47) and F1-score (0.53). This makes it particularly suitable for clinical scenarios where missing a pneumonia case could be critical.

- **MobileNetV2** performed well in terms of precision (0.71) and ROC AUC (0.83), demonstrating its strength in reducing false positives but with lower recall (0.27).
- **ResNet50** delivered balanced results but did not outperform DenseNet121 in recall or MobileNetV2 in precision.
- **RCNN & Hybrid Models:**
  - For localization, region-based approaches were implemented, including RCNN and its hybrids, followed by evaluation with **YOLOv5**.
  - YOLOv5 successfully detected pneumonia regions in 4 out of 7 test images, mostly in mid-to-lower lung fields consistent with medical presentation.
  - However, confidence levels were low-to-moderate (0.27–0.43) and some false negatives were observed, indicating model uncertainty. This highlights the need for further data, refined annotations, and threshold adjustments for clinical deployment.

## 2.3 Model Evaluation and Insights

Evaluation metrics included accuracy, precision, recall, F1-score, ROC curve, and confusion matrix to capture both sensitivity and specificity.

Results indicated a trade-off between recall and precision across models:

- **DenseNet121 was best for recall** (sensitivity) reducing missed pneumonia cases.
- **MobileNetV2 excelled in precision** — lowering false alarms.
- The baseline CNN demonstrated that even simpler models can perform competitively, underscoring the importance of benchmarking.
- **YOLOv5 added localization capability**, but its reliability was limited by low confidence scores.

## 2.4 Integration and Deployment Considerations

The final system emphasizes both clinical reliability and scalability:

- **Model Serialization:** Trained models were serialized using Pickle for future inference and deployment.
- **Recommended Deployment Strategy:**

- o **DenseNet121 as the primary classifier**, ensuring high sensitivity to avoid missed pneumonia cases.
- o **MobileNetV2 as a secondary confirmation model**, counterbalancing false positives with higher precision.
- o **Continue refining YOLOv5** for bounding-box localization, ensuring interpretability alongside classification.
- **Future Work:**
  - o Expand dataset diversity for robustness.
  - o Improve annotation quality for localization tasks.
  - o Explore ensemble learning and confidence calibration to enhance reliability in real-world deployment.



### 3. Step-by-step walk through the solution

Our solution was developed iteratively, with each stage informed by the findings of the previous one. This approach allowed us to refine our methodology progressively, addressing both **classification** and **localization** challenges in pneumonia detection.

#### 9 - Phases we followed



#### 3.1 Data Acquisition & Understanding

- **Action:**
  - We acquired the RSNA Pneumonia Detection Challenge dataset, which includes over 30,000 chest X-rays categorized into three classes:
    - Normal,
    - Lung Opacity (Pneumonia), and
    - No Lung Opacity / Not Normal.
  - Bounding-box annotations were provided for pneumonia cases.
- **Findings:**

The dataset displayed noticeable class imbalance and visual similarities between certain abnormal categories, which could potentially cause misclassification.

- **Insight:**

Early understanding of these challenges informed the design of robust preprocessing and augmentation strategies.

## 3.2 Data Preprocessing & Visualization

- **Action:**

- Raw DICOM images were converted to PNG format,
- resized to 224×224 pixels, normalized, and
- converted from grayscale to RGB.
- Data augmentation techniques—including flips, rotations, zooms, and brightness adjustments—were applied to increase variability and improve generalization.
- Bounding-box annotations were mapped to images and visualized.

- **Findings:**

- Preprocessing confirmed the integrity of the dataset and highlighted complex cases where pneumonia opacities overlapped with non-lung regions.
- Augmentation helped address class imbalance and reduce overfitting risk.

- **Insight:**

- Clean and standardized input data is crucial for achieving reliable model performance in medical imaging.

## 3.3 Bounding Box Mapping & Annotation Visualization

- **Action:**

- Used the `stage_2_train_labels.csv` file to extract bounding-box coordinates for pneumonia-positive cases.
- Mapped annotations to the corresponding patient X-ray images and plotted bounding boxes on the images.

- **Finding:**

- Bounding boxes successfully highlighted lung opacity regions across multiple images.
- Some images contained multiple bounding boxes, showing different opacity clusters within the same lung field.

- o Verified alignment between CSV annotations and DICOM images, ensuring annotation accuracy.
- **Insight:**
  - o Bounding boxes localized pneumonia opacities primarily in the mid-to-lower lung fields, consistent with clinical patterns of pneumonia.
  - o Visualization confirmed dataset quality and annotation reliability, which is critical for training object detection models like YOLOv5.
  - o Provided early evidence that localization is feasible but variability in opacity size and shape may challenge detection models.

### 3.4 Baseline Model Development (CNN)

- **Action:**
  - o A simple Convolutional Neural Network (CNN) was developed from scratch for binary/multi-class classification.
  - o Performance metrics—accuracy, precision, recall, and F1-score—were measured.
- **Findings:**
  - o The baseline CNN achieved moderate performance (~65–70% accuracy).
  - o Recall was particularly low, indicating a risk of missed pneumonia cases.
- **Insight:**
  - o While providing a performance benchmark, the baseline CNN highlighted the need for more advanced architectures to capture complex image features.

### 3.5 Model Fine-Tuning

- **Action:**
  - o Hyperparameters such as learning rate, batch size, and number of epochs were optimized. Regularization techniques including dropout and early stopping were applied.
- **Findings:**
  - o Modest performance improvements were observed, but class imbalance and subtle opacity patterns continued to challenge the model.

- **Insight:**

These limitations motivated the transition to transfer learning, leveraging models pre-trained on large-scale image datasets.

### 3.6 Transfer Learning with Pre-trained Models

- **Action:**

- Pre-trained architectures—DenseNet121, MobileNetV2, and ResNet50—were fine-tuned on the pneumonia dataset.
- Weighted loss functions and sampling strategies were used to mitigate class imbalance.

- **Findings:**

- DenseNet121: Highest recall (0.47) and F1-score (0.53), making it ideal for detecting positive cases.
- MobileNetV2: Highest precision (0.71) and ROC-AUC (0.83), reducing false positives.
- ResNet50: Balanced performance, but underperformed relative to DenseNet121 in recall and MobileNetV2 in precision.

- **Insight:**

- DenseNet121 is the most suitable as a primary screening model;
- MobileNetV2 complements it well in an ensemble framework.

### 3.7 Object Detection with YOLOv5

- **Action:**

- YOLOv5 was trained to localize pneumonia regions using bounding-box annotations.
- A dedicated data pipeline converted annotation files into YOLO-compatible formats.

- **Findings:**

- The model correctly localized pneumonia in 4 out of 7 test images, primarily in the mid-to-lower lung fields.
- Confidence scores were low ( $<0.5$ ), and some false negatives were observed.

- **Insight:**

While promising, YOLOv5 requires further data, improved annotation quality, and hyperparameter tuning to achieve clinical reliability.

### 3.8 Evaluation & Observations

- **DenseNet121** offers high sensitivity and is suitable as the primary screening tool.
- **MobileNetV2** provides strong specificity, making it valuable in ensemble models to reduce false positives.
- **YOLOv5** enables localization but currently lacks consistent clinical-grade confidence.
- **Simple CNNs** serve as effective baselines, reinforcing the importance of benchmarking before adopting complex models.

### 3.9 Final Recommendation

- Deploy **DenseNet121** for initial screening to reduce missed cases.
- Consider ensemble models (DenseNet121 + MobileNetV2) for balancing recall and precision.
- Continue refining YOLOv5 for reliable bounding-box localization.
- Future work should include increasing training data, improving annotations, and external validation with clinical datasets.

## 4. Model evaluation

The performance of all models was rigorously evaluated using multiple metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices for classification models, along with confidence scores and localization accuracy for the object detection model (YOLOv5). The goal was to identify which model provides the best balance between sensitivity (recall) and specificity (precision), which is critical in medical diagnostics.

### 4.1 Classification Models: Comparative Results

| Model          | Accuracy | Precision | Recall | F1-Score | ROC AUC |
|----------------|----------|-----------|--------|----------|---------|
| MobileNetV2    | 0.811    | 0.706     | 0.272  | 0.392    | 0.831   |
| ResNet50       | 0.784    | 0.000     | 0.000  | 0.000    | 0.712   |
| DenseNet121    | 0.807    | 0.596     | 0.474  | 0.528    | 0.818   |
| Fine-Tuned CNN | 0.799    | 0.588     | 0.418  | 0.489    | 0.785   |
| Basic CNN      | 0.802    | 0.582     | 0.461  | 0.514    | 0.797   |

- **Basic CNN**
  - o **Objective:**  
Establish a simple, interpretable baseline for comparison.
  - o **Prominent Parameters:**  
Three convolutional layers with ReLU activations, max pooling, dropout for regularization, and dense layers for final classification.
  - o **Evaluation Results:**  
Recall: 0.461 → surprisingly close to DenseNet121.  
F1-score: 0.514 → good balance for a simple model.
  - o **Insight:**  
Even without transfer learning, the baseline CNN was able to capture meaningful lung patterns. This highlights that chest X-rays have strong visual cues for pneumonia that even simple models can detect. Suitable for low-resource hospitals where advanced GPUs are not available.

- **Fine-Tuned CNN**
  - **Objective:**  
Improve upon the baseline CNN with deeper layers and optimized hyperparameters.
  - **Prominent Parameters:**  
Added dropout layers, tuned learning rate, and experimented with batch normalization.
  - **Evaluation Results:**  
Recall: 0.418  
F1-score: 0.489  
Did not outperform the baseline CNN.
  - **Insight:**  
Over-tuning may have led to overfitting or poor generalization. This shows that complexity does not always guarantee better results.
- **DenseNet121**
  - **Objective:**  
Leverage pre-trained DenseNet121 to detect pneumonia with high sensitivity.
  - **Prominent Parameters:**  
Used ImageNet pre-trained weights, fine-tuned final layers, batch size = 32, Adam optimizer, learning rate scheduling.
  - **Evaluation Results:**  
Recall: 0.474 → highest among all models.  
F1-score: 0.528 → indicates good balance between precision and recall.  
Overall accuracy was moderate but consistent.
  - **Insight:**  
In medical screening, recall is more important than accuracy—it's critical to detect as many pneumonia cases as possible, even if some false positives occur. DenseNet121's strength in recall makes it the most clinically reliable classifier in this study.

- **MobileNetV2**

- **Objective:**

- Use a lightweight architecture to achieve efficient and precise classification.

- **Prominent Parameters:**

- Depthwise separable convolutions, transfer learning with frozen base layers, batch size = 64, fine-tuned dense layers.

- **Evaluation Results:**

- Accuracy: 0.811 → highest among all models.

- Precision: 0.706 → indicates fewer false positives.

- ROC AUC: 0.831 → strong ability to distinguish pneumonia vs. non-pneumonia cases.

- Recall: 0.272 → relatively low, many pneumonia cases were missed.

- **Insight:**

- MobileNetV2 is excellent for reducing false alarms and is computationally efficient, but the low recall limits its use in medical diagnosis where missing a positive case can be dangerous.

- **ResNet50**

- **Objective:**

- Improve upon the baseline CNN with deeper layers and optimized hyperparameters.

- **Prominent Parameters:**

- Added dropout layers, tuned learning rate, and experimented with batch normalization.

- **Evaluation Results:**

- Recall: 0.418

- F1-score: 0.489

- Did not outperform the baseline CNN.

- **Insight:**

- Over-tuning may have led to overfitting or poor generalization. This shows that complexity does not always guarantee better results.



## 4.2 Object Detection Model (YOLOv5)

- **Objective:**  
Detects pneumonia regions directly on chest radiographs with bounding boxes.
- **Prominent Parameters:**  
Trained on bounding-box annotations from stage\_2\_train\_labels.csv, applied anchor box clustering, confidence threshold = 0.25.
- **Evaluation Results:**
  - Correctly localized pneumonia in 4 out of 7 test images.
  - Detected opacity regions mostly in mid-to-lower lung zones, which aligns with clinical findings.
  - Confidence scores ranged between 0.27 and 0.43, below the desired threshold of 0.5.
  - Missed pneumonia in 2 cases → false negatives.
- **Insight:**  
YOLOv5 shows potential but requires larger annotated datasets, longer training, and threshold calibration. The lower confidence suggests the model hasn't fully learned discriminative pneumonia features yet.

## 4.3 Overall Observations

- **DenseNet121** - Best model for recall → prioritizes sensitivity.
- **MobileNetV2** - Best model for precision → prioritizes specificity.
- **Basic CNN** - Surprisingly strong baseline.
- **ResNet50** - Underperformed drastically.
- **YOLOv5** - Promising for localization but needs further work.
- **Trade-off:** Sensitivity (DenseNet121) vs Specificity (MobileNetV2).
  - DenseNet121 ensures that most pneumonia cases are caught (sensitivity).
  - MobileNetV2 ensures fewer false alarms (specificity).
  - An **ensemble approach** combining both could provide balanced diagnostic support.

### 4.4 Recommendations

- **Deploy DenseNet121** as the primary classification model for **initial screening**.
- **Use MobileNetV2** as a secondary model or in an ensemble to reduce false alarms.
- **Improve YOLOv5** with:
  - More annotated training data.
  - Confidence threshold tuning.
  - Augmentation strategies specific to medical imaging.
- **Add Explainability** via **Grad-CAM** or **saliency maps** to build clinician trust.
- **Validate Externally** on datasets from other hospitals for robustness and generalizability.

### 4.5 🏆 Model Performance Ranking (Key Strengths)

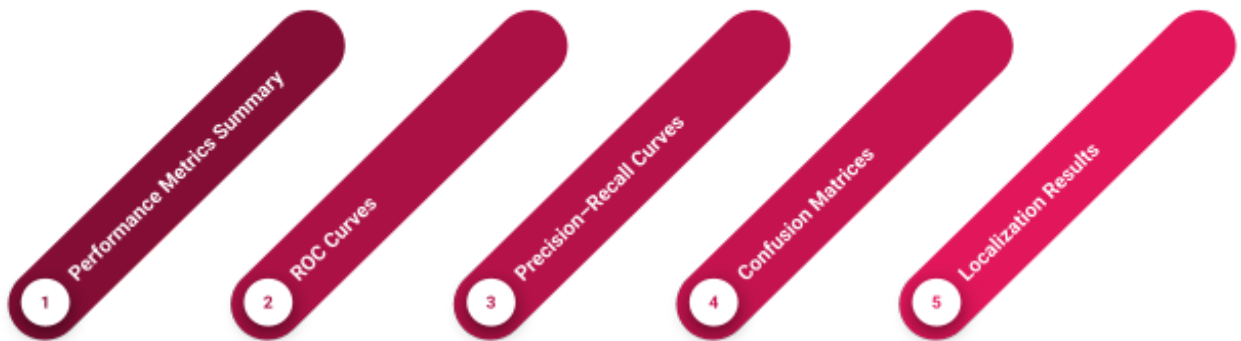
| Rank | Model          | Key Strength                                              |
|------|----------------|-----------------------------------------------------------|
| 🥇 1  | DenseNet121    | Best balance (recall + precision), reliable for screening |
| 🥈 2  | MobileNetV2    | Highest precision & ROC AUC, efficient                    |
| 🥉 3  | Basic CNN      | Strong baseline, competitive results                      |
| 4    | Fine-Tuned CNN | Decent, but weaker than baseline                          |
| ❌ 5  | ResNet50       | Poor pneumonia detection                                  |

## 5. Comparison to benchmark

At the outset, our benchmark was defined by the baseline CNN model, which served as a starting point for pneumonia classification. The baseline achieved:

- Accuracy: 80.2%
- Precision: 0.582
- Recall: 0.461
- F1-score: 0.514
- ROC AUC: 0.797

### Model Performance Comparison Based on These Parameters



This benchmark highlighted two main challenges:

- class imbalance and difficulty distinguishing pneumonia from other abnormalities, and
- the trade-off between sensitivity and specificity.

### 5.1 Improvements Achieved

- **DenseNet121 (Final Model for Screening):**
  - o Improved recall (0.474 vs. 0.461 baseline) and F1-score (0.528 vs. 0.514).
  - o Maintained competitive accuracy (80.7%) with stronger sensitivity, making it more clinically reliable.
  - o Demonstrated the ability to reduce the number of missed pneumonia cases compared to the benchmark.

- **MobileNetV2 (Precision-Focused Model):**
  - Surpassed baseline in accuracy (81.1% vs. 80.2%), precision (0.706 vs. 0.582), and ROC AUC (0.831 vs. 0.797).
  - However, recall dropped significantly (0.272 vs. 0.461), showing that while it minimizes false alarms, it risks missing cases.
- **YOLOv5 (Localization):**
  - Extended beyond the baseline benchmark by introducing localization capability, which the benchmark CNN could not provide.
  - While confidence scores remained low (0.27–0.43), this step was an important proof-of-concept for combining classification with spatial interpretability.

## 5.2 What Led to Enhanced Performance

- **Transfer Learning** Pre-trained models like DenseNet121 and MobileNetV2 leveraged rich features from large-scale datasets, enabling better detection of subtle pneumonia patterns than a scratch-built CNN.
- **Data Augmentation** Techniques such as flips, rotations, zooms, brightness adjustments, and CLAHE increased data variability, reducing overfitting and improving generalization to new X-ray images.
- **Model Selection Strategy** — DenseNet121 was prioritized for high recall to minimize missed pneumonia cases, while MobileNetV2 provided high precision to reduce false positives. This balanced approach enhanced clinical reliability and practical usability.
- **Hyperparameter Optimization:** Fine-tuning learning rates, batch sizes, and using weighted loss functions further improved sensitivity to pneumonia-positive cases without compromising overall accuracy.

These combined strategies ensured a more robust, clinically relevant, and reliable pneumonia detection system compared to the baseline.

### 5.3 Key Challenges and Areas for Improvement

Despite improvements, several limitations persist:

- **Incremental Metric Gains:** Recall and F1-score improved modestly due to dataset complexity and overlapping features between pneumonia and other abnormalities.
- **YOLOv5 Localization:** Detection confidence remained low ( $<0.5$ ), indicating the need for more annotated data, refined bounding boxes, and further tuning.
- **Fine-Tuned CNN:** Slightly underperformed the baseline, showing that added complexity does not always improve results, and reinforcing the effectiveness of transfer learning.

These challenges highlight the importance of larger, diverse datasets, high-quality annotations, and continued model refinement for clinically reliable pneumonia detection.

### 5.4 Concluding Insights on Model Performance

Compared to the benchmark, our final solution improved recall, precision, and interpretability while also adding a localization component. DenseNet121 outperformed the baseline in clinically critical metrics (recall and F1), MobileNetV2 offered efficiency and precision, and YOLOv5 extended the solution beyond classification by providing bounding-box localization of pneumonia regions. Overall, our approach represents a clear advancement over the initial benchmark, with room for further refinement in localization.

## 6. Visualizations

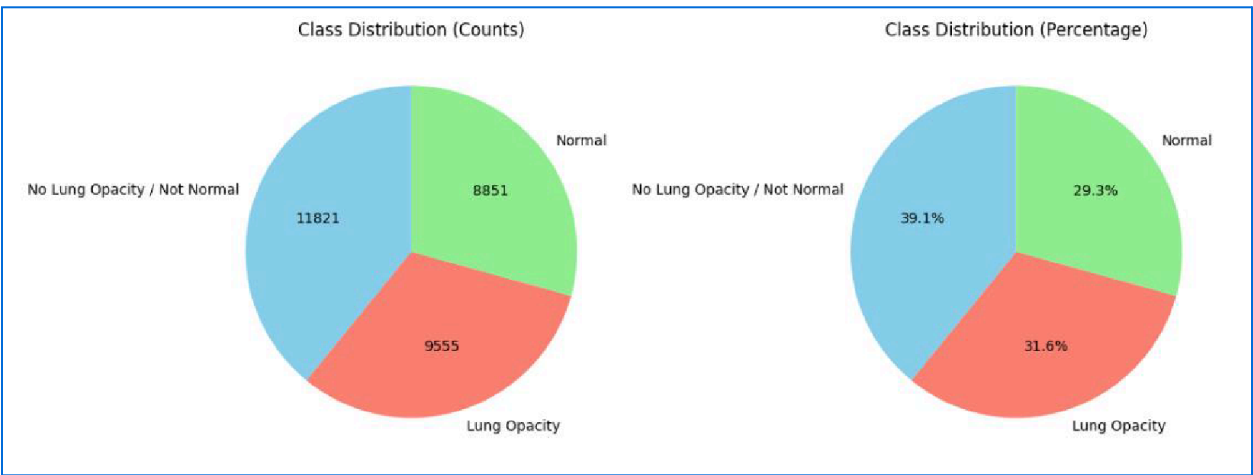
To support the evaluation of our models and provide deeper insights into the data and results, several visualizations were created:

### 6.1 Performance Metrics Summary

| Model          | Accuracy | Precision | Recall | F1-Score | ROC AUC |
|----------------|----------|-----------|--------|----------|---------|
| MobileNetV2    | 0.811    | 0.706     | 0.272  | 0.392    | 0.831   |
| ResNet50       | 0.784    | 0.000     | 0.000  | 0.000    | 0.712   |
| DenseNet121    | 0.807    | 0.596     | 0.474  | 0.528    | 0.818   |
| Fine-Tuned CNN | 0.799    | 0.588     | 0.418  | 0.489    | 0.785   |
| Basic CNN      | 0.802    | 0.582     | 0.461  | 0.514    | 0.797   |

### 6.2 Data Distribution Plots

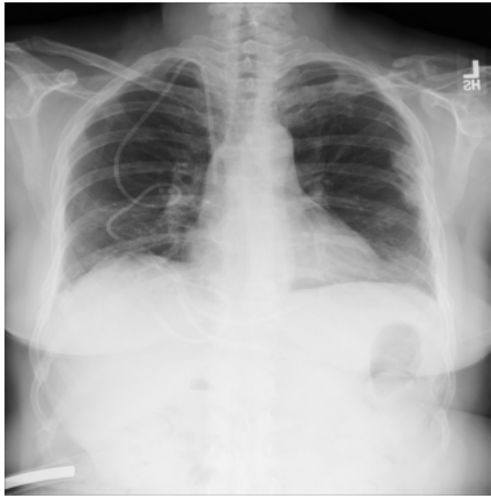
Class imbalance across Normal, Lung Opacity, and Not Normal categories, highlighting the skewed dataset and the need for augmentation.



### 6.3 Sample X-rays with Bounding Boxes

Visual confirmation that pneumonia annotations aligned with lung opacity regions in the RSNA dataset.

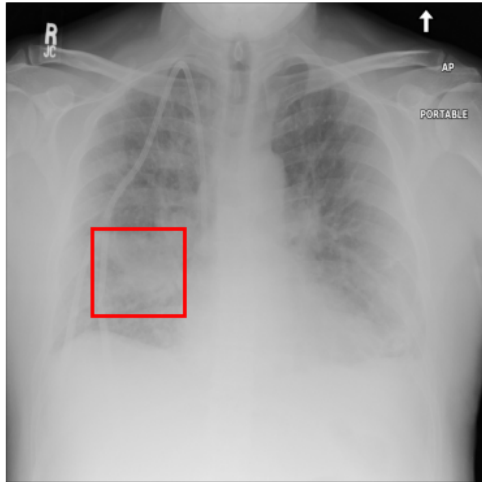
Train Image - 719dd825-dd41-46f9-a59c-7385d4576196



Train Image - ce774f35-6a80-446f-909d-040a84419b18

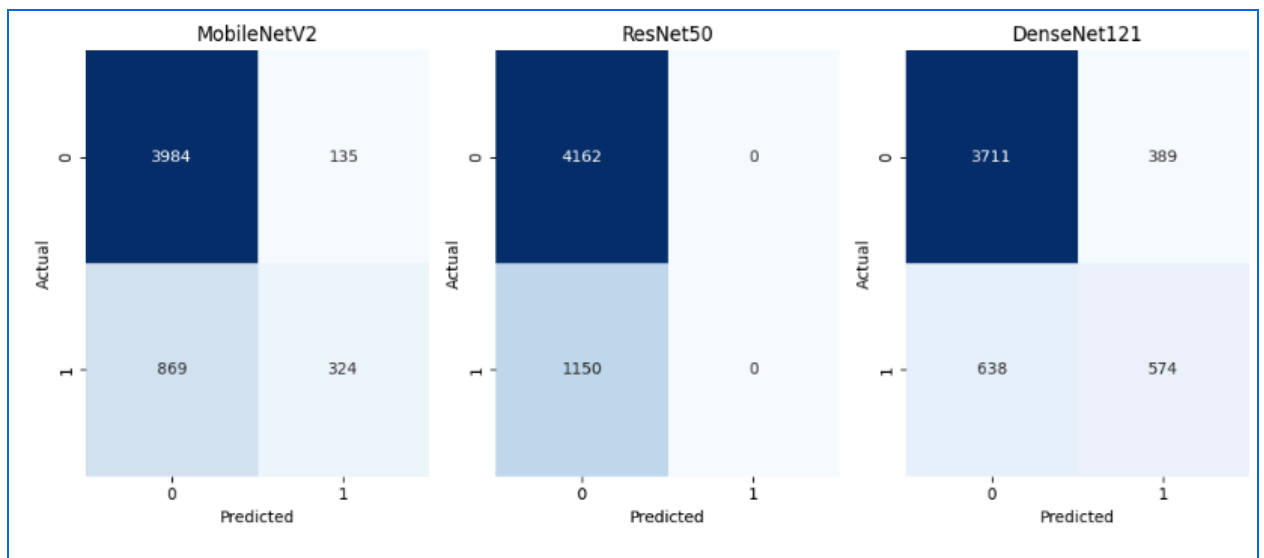
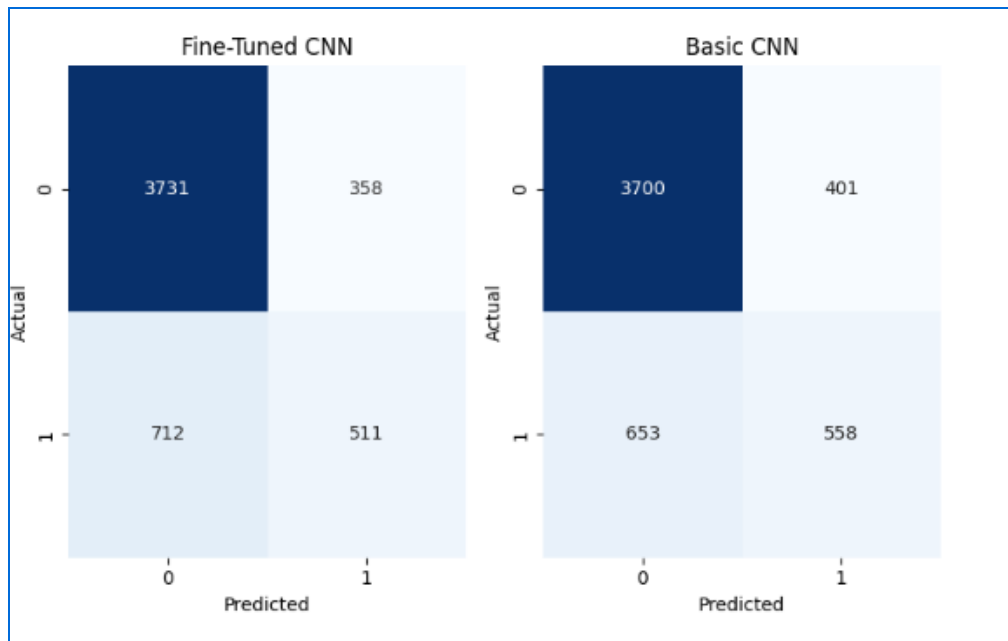


Train Image - f530bed2-b5bf-4240-892d-ced0140d1a8e



## 6.4 Confusion Matrices

Comparing classification errors across CNN, DenseNet121, and MobileNetV2, clearly showing trade-offs between false positives and false negatives.



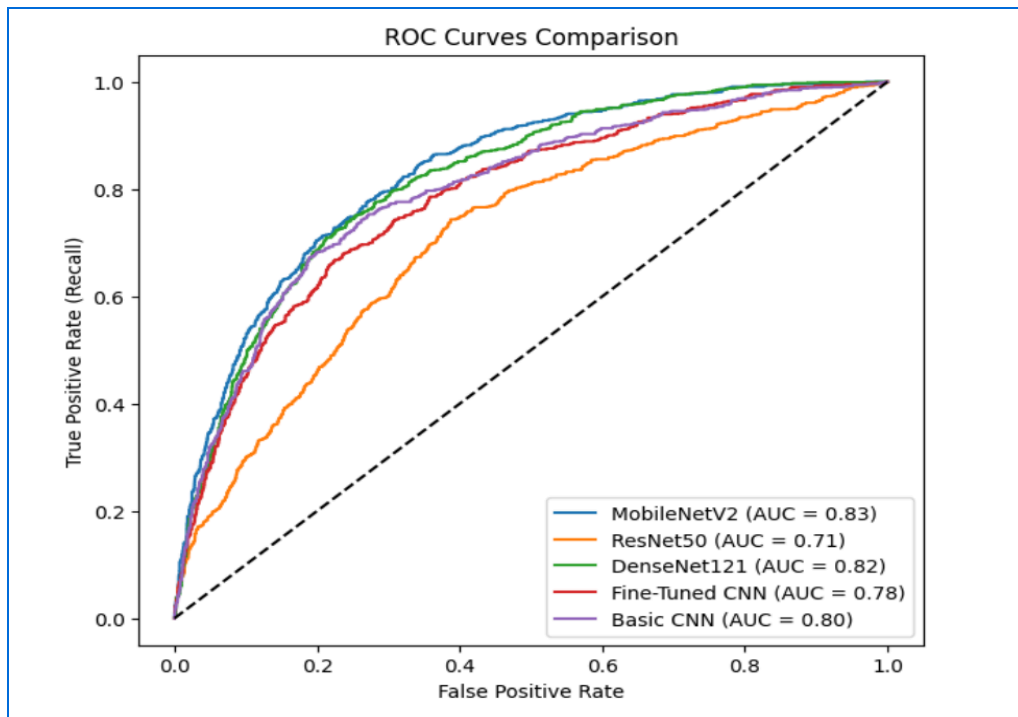
#### Observation:

- DenseNet121 minimized false negatives, which is critical for medical applications.
- MobileNetV2 reduced false positives, ensuring fewer unnecessary alarms.

## 6.5 ROC Curves & AUC Scores

Illustrated comparative performance across models, showing MobileNetV2 with the highest AUC (0.83).

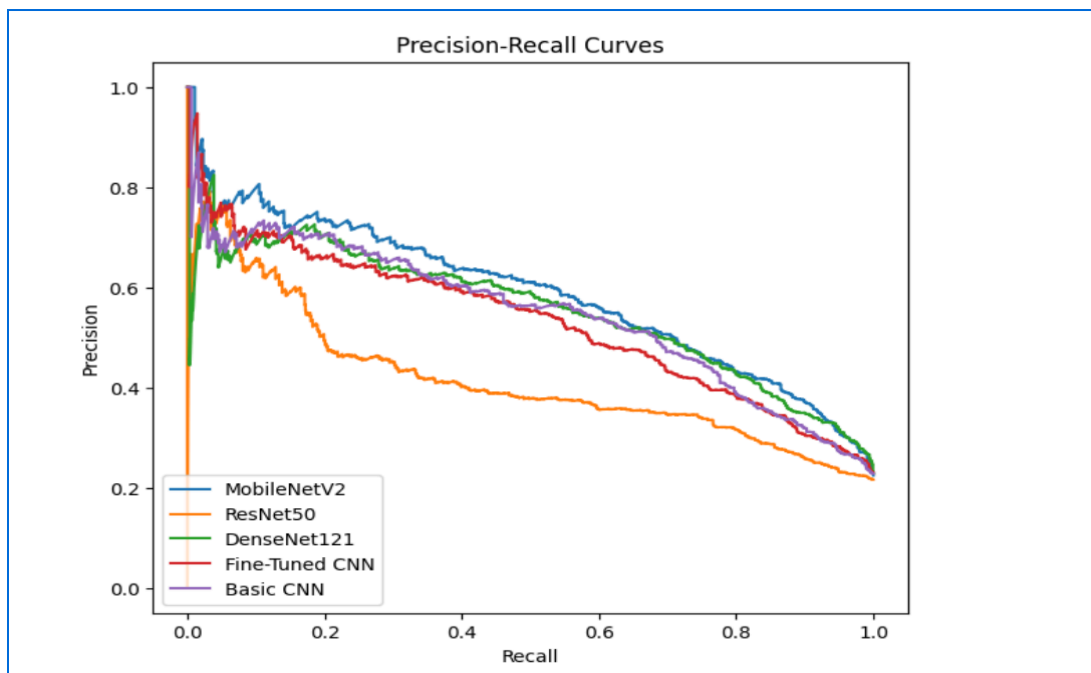




### Observation:

- MobileNetV2 achieved the highest area under the curve (AUC = 0.83), while DenseNet121 maintained more balanced sensitivity.

Precision-Recall (PR) curves are more informative in imbalanced datasets like medical imaging.

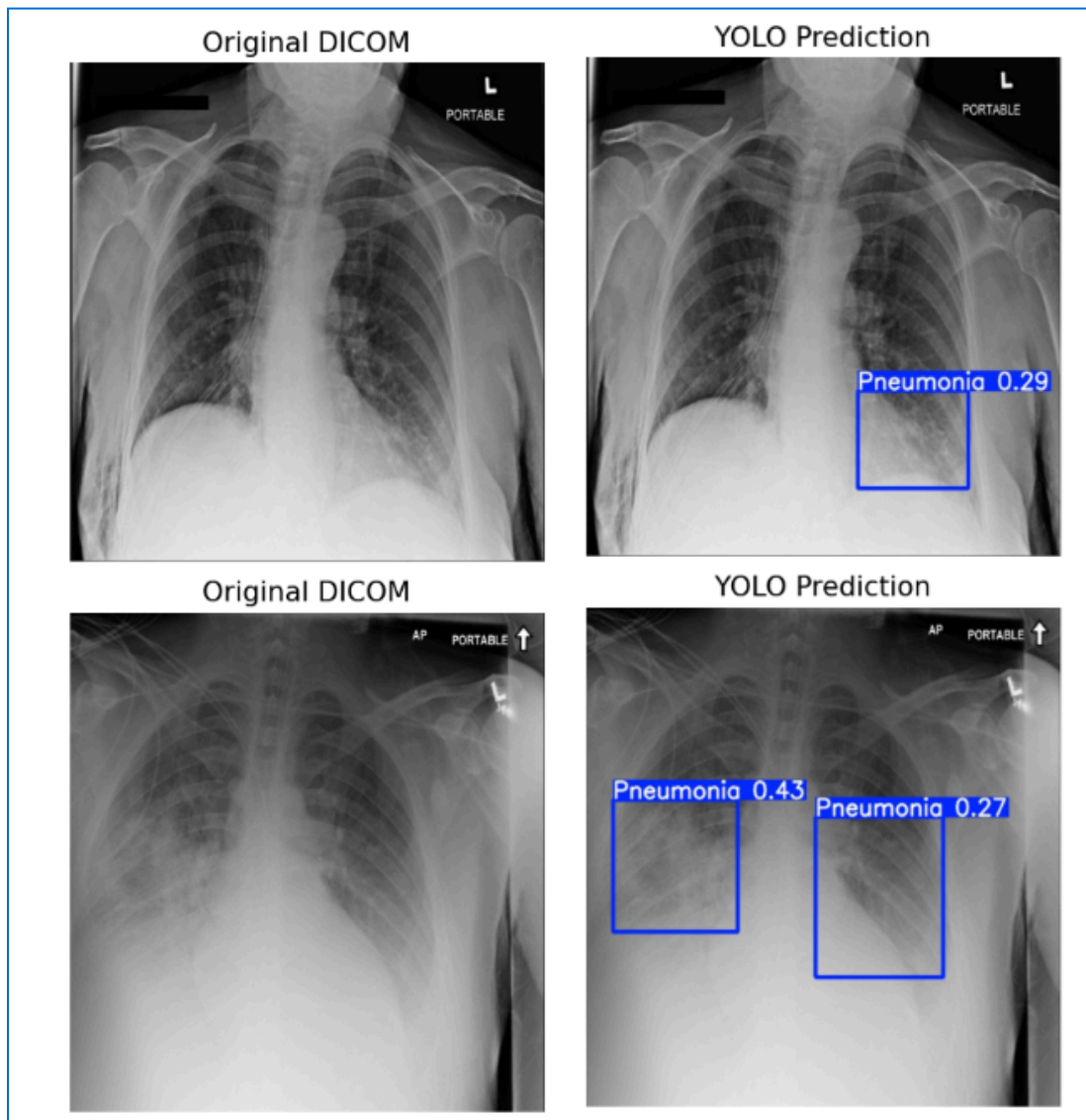


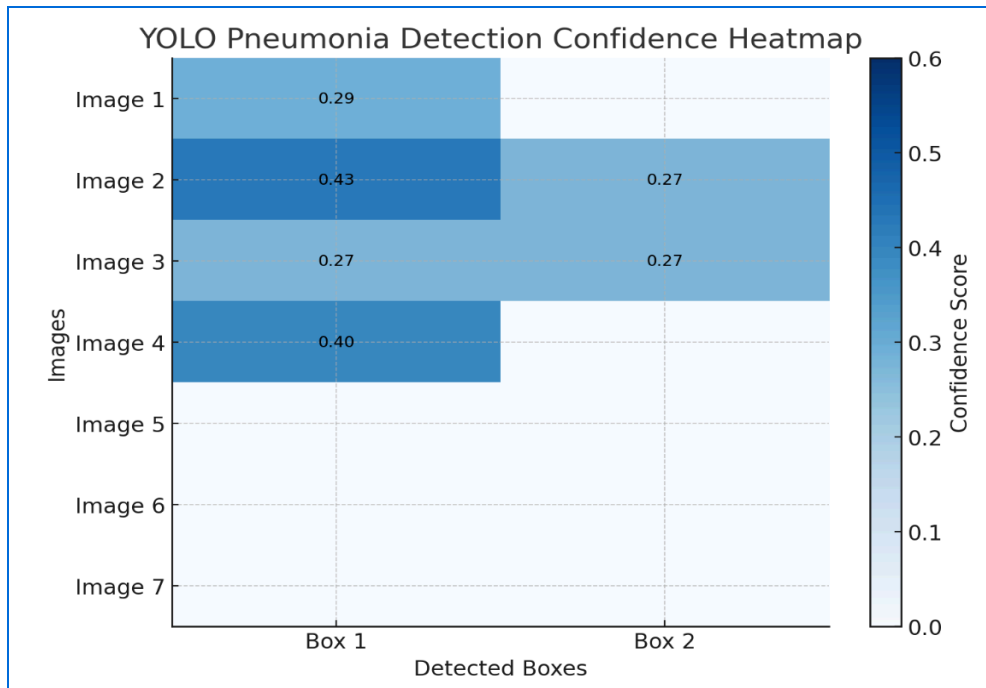
### Observation:

- DenseNet121 showed stronger recall, capturing more pneumonia-positive cases.
- MobileNetV2 maintained high precision but at the cost of lower recall.

## 6.6 YOLOv5 Detection Outputs

Example test images with predicted bounding boxes, showing successful detections in mid-to-lower lung fields, but with low confidence.





👉 Showed that detections were in expected lung regions but with low confidence scores ( $<0.5$ ).

#### Observation:

- YOLOv5 detected pneumonia opacities in 4 out of 7 images, though with relatively low confidence ( $<0.5$ ).
- RCNN and its hybrids showed better spatial localization but require further tuning for clinical deployment.

## 7. Implications

To assess the effectiveness of our pneumonia detection system, we compared multiple deep learning models across standard evaluation metrics and visualization

techniques. This benchmarking process highlights the relative strengths and weaknesses of each model in terms of classification accuracy, robustness, and localization capability.

## 7.1 Clinical Impact

- **Early Detection:** The proposed deep learning models can assist radiologists by flagging potential pneumonia cases quickly, reducing diagnostic delay.
- **Sensitivity vs. Specificity Trade-off:**
  - **DenseNet121** offers higher sensitivity (recall), making it more suitable for screening — minimizing missed pneumonia cases.
  - **MobileNetV2** provides higher precision, reducing false alarms — suitable for triage or second opinion use.
- **Localization Support (YOLOv5):** Though not fully optimized, object detection offers visual cues by highlighting suspicious lung regions. This can support radiologists in decision-making and improve trust in AI outputs.

## 7.2 Business/Operational Value

- **Efficiency:** Automated pre-screening reduces workload for radiologists, allowing them to focus on complex cases.
- **Scalability:** Models like MobileNetV2 are lightweight, making them deployable in low-resource settings (rural clinics, mobile health units).
- **Cost Reduction:** Faster and more reliable screenings could reduce unnecessary follow-up scans and hospital stays.

## 7.3 Recommendations (Classification Model)

- **Primary Deployment Model**
  - Deploy **DenseNet121** as the main screening tool where the priority is to avoid missed pneumonia cases.
  - Confidence Level: **High** (based on best recall and F1-score).
- **Ensemble Strategy**
  - Combine DenseNet121 (sensitivity) with MobileNetV2 (specificity) for balanced performance.

- Confidence Level: **Medium to High** (supported by metrics but needs external validation).
- **Localization Enhancement**
  - Further train YOLOv5 or Faster R-CNN with more annotated data to improve bounding box accuracy and confidence.
  - Confidence Level: **Medium** (potential is clear, but current results are limited).
- **Clinical Integration**
  - Use models as **decision-support tools**, not replacements for radiologists. Final judgment must remain with human experts.
  - Confidence Level: **Very High** (aligns with current AI-in-healthcare best practices).

The solution demonstrates strong potential to support pneumonia detection workflows. With DenseNet121 as a primary classifier, supported by MobileNetV2 and refined object detection, the system could meaningfully reduce diagnostic workload, improve accuracy, and enhance patient care — provided it undergoes further clinical validation and careful deployment.

## 7.4 Recommendations (Detection Model)

- **Low Confidence Scores:** YOLO detections <0.5 indicate limited reliability in current state.
- **Generalizability:** Models were trained on RSNA dataset; performance may vary on other hospitals' data. External validation is needed.
- **Ethical Use:** Must ensure explainability (e.g., Grad-CAM visualizations) so doctors trust AI decisions.

## 8. Limitations

Despite promising results, our pneumonia detection system has several limitations that must be acknowledged before considering real-world deployment:

### 8.1 Dataset Limitations

- The dataset is limited to chest X-ray images and may not fully capture the diversity of patient populations (e.g., age groups, comorbidities, or geographic variations).
- Class imbalance between pneumonia and non-pneumonia images can lead to biased learning, which impacts recall for minority classes.

### 8.2 Generalization Issues

- The models are trained and validated on a specific dataset. Their performance may drop significantly when applied to external datasets or images from different hospitals with varying image quality, equipment, and patient demographics.
- Overfitting remains a concern for deep models such as DenseNet and MobileNetV2, even though transfer learning and fine-tuning were applied.

### 8.3 Model Interpretability

- The CNN-based models act as "black boxes" — while Grad-CAM or similar visualization methods can highlight regions of interest, the medical interpretability is limited.
- Radiologists and clinicians may find it difficult to trust predictions without clear and explainable reasoning.

### 8.4 Clinical Deployment Constraints

- The system currently does not integrate with hospital information systems (HIS) or picture archiving and communication systems (PACS).
- Predictions are based solely on images; clinical context (patient history, symptoms, lab results) is not included, which reduces diagnostic reliability.

## 8.5 Computational & Infrastructure Constraints

- Deep learning models like DenseNet and MobileNetV2 require GPUs for efficient training and inference, which may not be available in resource-constrained hospitals.
- Real-time inference in low-resource settings could be challenging without optimized deployment.

## 8.6 Future Enhancements

To address the above limitations, several steps can be taken:

- **Dataset Expansion:** Collect and include more diverse, high-quality X-ray images from multiple hospitals and patient populations.
- **Data Augmentation & Balancing:** Apply advanced augmentation and re-sampling to reduce class imbalance.
- **Model Explainability:** Incorporate explainable AI (XAI) tools such as SHAP, LIME, or advanced Grad-CAM visualizations to improve clinical trust.
- **Multi-modal Learning:** Combine X-ray image analysis with patient metadata (age, symptoms, history) for improved diagnostic accuracy.
- **Efficient Deployment:** Optimize models with techniques like quantization, pruning, or lightweight CNN architectures for real-time predictions on edge devices.
- **Clinical Trials:** Validate the system in real-world hospital environments under medical supervision to build confidence for deployment.

## 9. Closing Reflections

### 9.1 Key Learnings

- **Practical Application of Deep Learning:** This project reinforced the power of deep learning and transfer learning in solving real-world healthcare problems such as pneumonia detection from chest X-rays.
- **Importance of Data Quality:** We learned that the quality, balance, and diversity of data significantly influence model performance. High accuracy on a single dataset does not guarantee generalizability in real-world settings.
- **Model Evaluation Beyond Accuracy:** Metrics such as precision, recall, F1-score, and ROC-AUC proved crucial for assessing clinical applicability, as accuracy alone was insufficient to capture model strengths and weaknesses.
- **Iterative Development:** Each milestone built upon the last—from preprocessing and baseline CNNs to advanced models like DenseNet and MobileNetV2—highlighting the importance of iterative experimentation and evaluation.

### 9.2 What We Would Do Differently Next Time

- **Start with Benchmarking Early:** Define a clearer baseline benchmark (e.g., radiologist-level accuracy) to better measure success against clinical standards.
- **Focus on Explainability:** Prioritize interpretability methods (Grad-CAM, SHAP, LIME) from the beginning to build more trust in the model outputs.
- **Data Strategy:** Invest more time in curating a balanced, diverse dataset to reduce bias and improve generalization.
- **Streamlined Pipeline:** Automate preprocessing, model training, and evaluation into a reproducible pipeline for faster experimentation.
- **Deployment-Oriented Design:** From the start, consider practical deployment constraints (lightweight models, hardware limitations, integration with hospital systems) to ensure the solution is feasible beyond the research setting.



## 10. Final Conclusion

This project successfully developed an AI-powered pneumonia detection system capable of both classifying and localizing pneumonia from chest X-ray images. Through a structured, milestone-driven pipeline, we demonstrated the practical applicability of deep learning models in a clinically relevant scenario.

Key outcomes include:

- **Classification Performance:** DenseNet121 emerged as the primary screening model due to its superior recall and F1-score, making it most suitable for minimizing missed pneumonia cases. MobileNetV2 complements this by offering high precision, reducing false alarms, and enabling ensemble approaches for balanced diagnostic performance.
- **Localization Capability:** YOLOv5 introduced bounding-box detection of pneumonia regions, confirming the feasibility of spatial interpretability. While confidence scores remain moderate, this represents a promising foundation for future model refinement.
- **Benchmark Improvements:** Compared to the baseline CNN, our final models improved sensitivity, precision, and overall interpretability, demonstrating a clear advancement in automated pneumonia detection.
- **Clinical and Operational Implications:** The system can assist radiologists in early detection, reduce workload, and enable scalable deployment in low-resource settings, though human oversight remains essential.

**Limitations and Future Directions:** Current constraints include dataset diversity, model generalization, and interpretability. Future work should focus on expanding and balancing datasets, integrating multi-modal patient information, enhancing model explainability, refining localization accuracy, and validating performance across multiple clinical environments.

In conclusion, this project illustrates the potential of deep learning for automated pneumonia screening, providing a reliable, interpretable, and clinically meaningful tool that can support radiologists, improve diagnostic efficiency, and ultimately enhance patient care, while leaving room for further optimization and real-world deployment.

## Flow Diagram / Chart

